Complete Interview Preparation Resource List (4–5 Day Plan)

1. Kubernetes Fundamentals (Must Know)

Watch First

1. Kubernetes Crash Course – TechWorld with Nana https://youtu.be/X48VuDVv0do

Read

1. Kubernetes Concepts https://kubernetes.io/docs/concepts/

Focus only on:

- Pods
- Deployments
- ReplicaSets
- Services
- Namespaces
- ConfigMaps
- Secrets
- Ingress

Must Understand

- Pod lifecycle
- CrashLoopBackOff
- ImagePullBackOff
- Liveness Probe
- Readiness Probe
- Resource Requests & Limits

Hands-On Commands

```
kubectl get pods
kubectl describe pod
kubectl logs
kubectl exec
kubectl get events
kubectl get deployments
kubectl rollout history
kubectl rollout undo
```

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2. Kubernetes Controllers & CRDs (Most Important)

Watch

1. Kubernetes Operators Explained https://youtu.be/i9V4oCa5f9I

2. Controller Runtime Tutorial https://youtu.be/1j4P1ddvN7A

Read

1. Kubernetes Operator Pattern https://kubernetes.io/docs/concepts/extend-kubernetes/operator/

2. Kubebuilder Book

https://book.kubebuilder.io

Read only:

- What is an Operator?
- CRDs
- Reconciliation Loop
- Controller Runtime

Must Know

CRD

- Why use CRDs?
- Spec vs Status

Reconciliation Loop

```
Watch Event
↓
Fetch Desired State
↓
Fetch Actual State
↓
Compare
↓
Take Action
↓
Update Status
```

Finalizers

Know:

- Why needed
- Deletion lifecycle

Owner References

Know:

- Garbage collection
- Parent-child relationships

Status Subresource

Know:

- Spec = Desired
- Status = Current

ResourceVersion

Know:

- Optimistic concurrency control
- Conflict handling

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3. Golang (Very Important)

Learn

Go By Example

https://gobyexample.com

Read:

- Structs
- Interfaces
- Methods
- Goroutines
- Channels
- Mutex
- WaitGroups

- Context

Watch

Go Tutorial

https://youtu.be/yyUHQIec83I

Concurrency

https://youtu.be/LvgVSSpwND8

Must Know Questions

- Interface vs Struct
- Pointer vs Value Receiver
- Goroutines
- Channels
- Mutex
- Context Cancellation
- Select Statement
- Error Handling

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4. Linux Fundamentals

Learn

Linux Journey

https://linuxjourney.com

Read:

- Processes
- Permissions
- Networking
- File System

Commands

```
ps
top
htop
grep
awk
```

```
sed
curl
find
chmod
chown
netstat
ss
tail
less
```

Practice

Know how to:

- Find a process
- Kill a process
- Check memory
- Check disk usage
- Check open ports
- View logs

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5. Docker

Watch

Docker Tutorial

https://youtu.be/3c-iBn73dDE

Must Know

- Image
- Container
- Dockerfile
- Volumes
- Networking
- Multi-stage Builds

Commands

```
docker build
docker run
docker ps
docker logs
```

```
docker exec
docker images
```

Common Questions

- Docker vs VM
- What is a layer?
- What happens during docker build?

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6. Distributed Systems

Read

System Design Primer

https://github.com/donnemartin/system-design-primer

Watch

CAP Theorem

https://youtu.be/k-Yaq8AHlFA

Learn

- CAP Theorem
- Eventual Consistency
- Idempotency
- Retry Mechanisms
- Exponential Backoff
- Distributed Tracing

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7. CI/CD

Read

GitHub Actions

https://docs.github.com/en/actions

Learn

Pipeline Flow

```
Code Push
↓
Build
↓
Test
↓
Lint
↓
Container Build
↓
Deploy
```

Know

- Blue-Green Deployment
- Rolling Deployment
- Canary Deployment

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8. Coding Interview Preparation

NeetCode

https://neetcode.io

Practice These

Arrays

- Two Sum
- Best Time To Buy Stock

Sliding Window

- Longest Substring Without Repeating Characters

Binary Search

- Binary Search
- Search Insert Position

Trees

- Invert Binary Tree
- Level Order Traversal

Graphs

- Number Of Islands

Heap

- Top K Frequent Elements

Intervals

- Merge Intervals

Goal

15–20 Medium Problems

Don't attempt Hard problems.

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9. Mock Interview Questions

Prepare Strong Answers For

Kubernetes

- What is a CRD?
- Explain reconciliation.
- What are finalizers?
- What are owner references?
- What is a status subresource?
- How would you debug CrashLoopBackOff?

Golang

- Goroutines vs Threads?
- Channel vs Mutex?
- Context package?
- Interfaces?

Linux

- How do you check open ports?
- How do you view logs?

- How do you find a process?

Resume

- Tell me about the controller you built.
- Explain the 40% latency reduction.
- Hardest production bug.
- Why Go?
- Why Kubernetes?

Best YouTube Channels

1. TechWorld with Nana
2. ByteByteGo
3. Hussein Nasser
4. Ardan Labs
5. JustForFunc (Go)
6. KodeKloud
7. Gaurav Sen

If You Have Only One Day Left

Priority Order:

1. Kubernetes Controllers & CRDs
2. Golang Concurrency
3. Linux Basics
4. Docker
5. Coding Questions
6. Distributed Systems
7. CI/CD